

Erdős Problem #1133

EP-1133

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Let $C > 0$. There exists $\epsilon > 0$ such that if n is sufficiently large the following holds.

For any $x_1, \dots, x_n \in [-1, 1]$ there exist $y_1, \dots, y_n \in [-1, 1]$ such that, if P is a polynomial of degree $m < (1 + \epsilon)n$ with $P(x_i) = y_i$ for at least $(1 - \epsilon)n$ many $1 \leq i \leq n$, then $\max_{x \in [-1, 1]} |P(x)| > C$.